Himaayah Schools
ACADEMY

GUIDE

وَأَنزَلْنَا الْحَدِيدَ فِيهِ بَأْسٌ شَدِيدٌ وَمَنَافِعُ لِلنَّاسِ

"And We sent down iron, in which is great strength and benefits for people." (al-Ḥadīd 57:25)

JAMB Chemistry

A UTME study guide: the syllabus, mole calculations, key facts and tests, worked examples and a practice test.

بِسْمِ اللَّهِ الرَّحْمٰنِ الرَّحِيمِ

Before you begin

UTME Chemistry has two kinds of question: calculations, almost all built on the mole, and facts (definitions, tests, properties, reactions and organic chemistry). Students who master mole calculations and memorise the key facts score well. This guide gives you the syllabus in six areas, the formulas and facts to know, nine worked examples, common traps, an 8-week plan and a practice test.

The UTME at a glance	
Subjects	Use of English and three subjects for your course. Chemistry is required for many science, medical and engineering courses.
Questions	Use of English 60; each other subject 40; 180 questions in all, on computer (CBT).
Time	About two hours for all four papers, so roughly 40 seconds a question. You move freely between subjects.
Marking	Each subject is scored out of 100, for a total out of 400. There is no negative marking: never leave a question blank.

Check the official details

Exam rules and syllabuses can change from year to year. Always check the current JAMB brochure and the official Chemistry syllabus on the JAMB website or e-facility portal, and follow them where they differ from this guide.

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Part 1 The syllabus at a glance

Study from the **JAMB syllabus**, not only from your school notes: every UTME question comes from a topic on it. The topics fall into these areas:

Area	Topics
Matter and the mole	Separation of mixtures and purification; laws of chemical combination; chemical formulae and equations; the mole concept and stoichiometry; kinetic theory and the gas laws
Atoms, bonds and the periodic table	Atomic structure, isotopes and electronic configuration; radioactivity and nuclear reactions; the periodic table and periodic trends; chemical bonding: ionic, covalent, dative, metallic, hydrogen bonds; shapes of molecules
Air, water and solutions	Composition of air; water: hardness, treatment, water of crystallisation; solubility and solubility curves; environmental pollution
Reactions	Acids, bases and salts, pH and indicators; oxidation and reduction, oxidation numbers; electrolysis; energy changes; rates of reaction; chemical equilibrium and Le Chatelier's principle
Inorganic chemistry	Non-metals and their compounds: hydrogen, halogens, oxygen, sulphur, nitrogen, carbon; metals and their compounds: sodium, calcium, aluminium, tin, iron, copper; extraction of metals; alloys
Organic chemistry and industry	Hydrocarbons: alkanes, alkenes, alkynes, aromatics; alcohols, acids, esters, amines; isomerism and IUPAC naming; petroleum, polymers, fats, soaps and detergents; carbohydrates and proteins; chemical industries

Mole calculations, gas laws and electrolysis appear every year and are worth mastering early. Organic chemistry and the tests for ions and gases are the facts most often tested.

Part 2 Formulas and key facts

The mole and solutions

Quantity	Formula	Notes
Amount from mass	n = mass ÷ molar mass	mol, g, g/mol
Amount from gas volume	n = volume ÷ 22.4 dm^3 (at s.t.p.)	s.t.p. = 273 K, 1 atm
Number of particles	$N = n \times 6.02 \times 10^{23}$	Avogadro's constant
Concentration	$c = n \div V(dm^3)$; $g/dm^3 = mol/dm^3 \times$ molar mass	1 dm^3 = 1000 cm^3
Titration	$c_1V_1/n_1 = c_2V_2/n_2$ (n = mole ratio from the equation)	acid–base
Percentage composition	% of element = (mass of element in formula ÷ molar mass) × 100	—
Empirical formula	Divide % by relative atomic mass, then by the smallest result	then scale to M

Gases, acids and electrolysis

Quantity	Formula	Notes
Gas laws	$P_1V_1/T_1 = P_2V_2/T_2$ (T in kelvin)	K = °C + 273
Graham's law	$rate_1/rate_2 = \sqrt{(M_2/M_1)}$	lighter gases diffuse faster
pH	$pH = -\log[H^+]$; pH + pOH = 14	<7 acid, 7 neutral, >7 alkaline
Charge	Q = It; 1 faraday = 96 500 C = 1 mole of electrons	C, A, s
Electrolysis	moles of substance = Q ÷ (96 500 × charge on ion)	Cu^{2+} needs 2 F per mole
Energy	ΔH negative: exothermic; ΔH positive: endothermic	kJ/mol

Tests to memorise

Test for	Method	Positive result
Hydrogen	Lighted splint	Burns with a "pop"
Oxygen	Glowing splint	Relights
Carbon dioxide	Bubble through limewater	Turns milky
Ammonia	Damp red litmus	Turns blue; white fumes with HCl
Chlorine	Damp blue litmus	Turns red, then bleached
Sulphur dioxide	Acidified $K_2Cr_2O_7$ paper	Orange to green
Hydrogen chloride	Ammonia (glass rod)	Dense white fumes
Water	Anhydrous $CuSO_4$	White to blue
Chloride ions	$AgNO_3$ with dilute HNO_3	White precipitate
Sulphate ions	$BaCl_2$ with dilute HCl	White precipitate
Unsaturation (C=C)	Bromine water	Decolourised

Organic series

Series	General formula	Example
Alkanes	C_nH_{2n+2}	Methane CH_4, ethane C_2H_6
Alkenes	C_nH_{2n}	Ethene C_2H_4
Alkynes	C_nH_{2n-2}	Ethyne C_2H_2
Alkanols	$C_nH_{2n+1}OH$	Ethanol C_2H_5OH
Alkanoic acids	$C_nH_{2n+1}COOH$	Ethanoic acid CH_3COOH
Esters	RCOOR′	Ethyl ethanoate, from ethanol + ethanoic acid

Useful relative atomic masses: H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, Al = 27, S = 32, Cl = 35.5, K = 39, Ca = 40, Fe = 56, Cu = 64, Zn = 65. JAMB usually gives the values you need in the question.

Part 3 Worked examples

Example 1. How many moles are in 4.4 g of carbon dioxide? What volume does it occupy at s.t.p.? (C = 12, O = 16)

→ Molar mass of CO_2 = 12 + 32 = 44 g/mol

→ n = 4.4 ÷ 44 = 0.1 mol

→ Volume = 0.1 × 22.4 = 2.24 dm^3

Answer: 0.1 mol; 2.24 dm^3

Example 2. What mass of calcium oxide is obtained by heating 50 g of calcium carbonate? (Ca = 40, C = 12, O = 16)

→ $CaCO_3 \rightarrow CaO + CO_2$

→ 100 g of $CaCO_3$ gives 56 g of CaO

→ 50 g gives 56 × 50/100 = 28 g

Answer: 28 g

Example 3. 4 g of sodium hydroxide is dissolved to make 500 cm^3 of solution. Find its concentration. (Na = 23, O = 16, H = 1)

→ Molar mass NaOH = 40 g/mol; n = 4 ÷ 40 = 0.1 mol

→ V = 500 cm^3 = 0.5 dm^3

→ c = 0.1 ÷ 0.5 = 0.2 mol/dm^3

Answer: 0.2 mol/dm^3

Example 4. 25.0 cm^3 of sodium hydroxide solution is neutralised by 20.0 cm^3 of 0.10 mol/dm^3 hydrochloric acid. Find the concentration of the alkali.

→ $NaOH + HCl \rightarrow NaCl + H_2O$ (1 : 1)

→ Moles of HCl = $0.10 \times 20.0/1000 = 0.002$ mol = moles of NaOH

→ $c = 0.002 \div 0.025 = 0.08$ mol/dm^3

Answer: 0.08 mol/dm^3

Example 5. What mass of copper is deposited when a current of 2 A flows through copper(II) sulphate solution for 965 s? (Cu = 64, F = 96 500 C)

→ $Q = It = 2 \times 965 = 1930$ C = 0.02 F

→ $Cu^{2+} + 2e^- \rightarrow Cu$: 2 F deposit 1 mol

→ Moles of Cu = 0.01; mass = $0.01 \times 64 = 0.64$ g

Answer: 0.64 g

Example 6. 200 cm^3 of a gas at 27 °C is heated to 127 °C at constant pressure. Find the new volume.

→ Charles' law: $V_1/T_1 = V_2/T_2$, with T in kelvin

→ $V_2 = 200 \times 400/300$

Answer: 266.7 cm^3 (about 267 cm^3)

Example 7. Find the pH of 0.01 mol/dm^3 hydrochloric acid.

→ HCl is a strong acid: $[H^+] = 0.01 = 10^{-2}$ mol/dm^3

→ $pH = -\log 10^{-2} = 2$

Answer: 2

Example 8. A compound contains 40% carbon, 6.7% hydrogen and 53.3% oxygen. Its molar mass is 180 g/mol. Find its molecular formula.

→ C: 40/12 = 3.33; H: 6.7/1 = 6.7; O: 53.3/16 = 3.33

→ Divide by 3.33: C 1, H 2, O 1, so the empirical formula is CH_2O (30 g/mol)

→ $180 \div 30 = 6$, so the molecular formula is $C_6H_{12}O_6$

Answer: $C_6H_{12}O_6$

Example 9. How many times faster does hydrogen diffuse than oxygen? (H = 1, O = 16)

→ $\text{rate}(H_2)/\text{rate}(O_2) = \sqrt{(M(O_2)/M(H_2))} = \sqrt{(32/2)}$

→ $= \sqrt{16} = 4$

Answer: 4 times faster

Part 4 Traps that cost marks

Trap	What to do
Forgetting to convert cm^3 to dm^3	Divide cm^3 by 1000 before using c = n/V.
Using °C in gas laws	Always convert to kelvin.

Ignoring the mole ratio	Balance the equation first, then use the ratio of moles, not of masses.
Diatomic gases	Hydrogen, oxygen, nitrogen and the halogens are H_2, O_2, N_2, Cl_2: use their molecular masses.
Electrolysis: charge on the ion	Al^{3+} needs 3 F per mole, Cu^{2+} needs 2 F, Na^+ and Ag^+ need 1 F.
Oxidation and reduction mixed up	OIL RIG: Oxidation Is Loss of electrons, Reduction Is Gain. Oxidation number goes up in oxidation.
Le Chatelier and catalysts	A catalyst speeds up both directions equally: it does not change the position of equilibrium.
Organic naming	Find the longest chain, number from the end nearer the functional group, and give the lowest numbers.
Not reading “which is NOT”	Underline NOT, EXCEPT and LEAST before you answer.

Part 5 An 8-week plan

Start at least eight weeks before the exam. Each week: learn the topics, make short notes and solve past questions under time. Keep one day a week for revision.

Week	Focus	Practice
1	Separation, laws of combination, formulae and equations, the mole	Mole calculations: 30 questions
2	Gas laws, kinetic theory, atomic structure, isotopes	Gas law and isotope problems
3	Periodic table, bonding, shapes; air and water	Mixed past questions
4	Solubility, acids, bases and salts, pH; pollution	Titration and pH calculations
5	Redox, electrolysis, energy changes, rates, equilibrium	Electrolysis calculations; Le Chatelier questions
6	Non-metals and their compounds; metals and extraction	Tests for gases and ions
7	Organic chemistry; industry; nuclear chemistry	Naming, isomerism and reactions
8	Revision of weak areas; full timed mock exams	Two full CBT mocks; review every error

Exam-day reminders

Read every option before choosing. Eliminate the clearly wrong ones first.

Do not spend more than a minute on one question: flag it and come back.

Answer every question; there is no penalty for a wrong answer.

Check that your answers are saved before you submit, and keep an eye on the clock for all four subjects.

Test Practice test

Give yourself 10 minutes for these 15 questions, as in the real exam. The answers follow.

1. Which method separates the components of crude oil?

a) Filtration b) Fractional distillation c) Crystallisation d) Sublimation

2. How many moles are in 9 g of water? (H = 1, O = 16)

a) 0.25 b) 2 c) 0.5 d) 1

3. The volume of 0.5 mol of a gas at s.t.p. is:

a) 22.4 dm^3 b) 5.6 dm^3 c) 44.8 dm^3 d) 11.2 dm^3

4. An element with electronic configuration 2, 8, 1 is in:

a) Group 1, period 3 b) Group 3, period 1 c) Group 8, period 2 d) Group 2, period 3

5. The bond in sodium chloride is:

a) Covalent b) Metallic c) Ionic d) Hydrogen

6. Which gas turns limewater milky?

a) Oxygen b) Hydrogen c) Ammonia d) Carbon dioxide

7. A solution of pH 12 is:

a) Strongly alkaline b) Strongly acidic c) Neutral d) Weakly acidic

8. In the reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, zinc is:

a) Reduced b) Oxidised c) A catalyst d) Unchanged

9. During electrolysis, cations move to the:

a) Anode b) Electrolyte surface c) Cathode d) Salt bridge

10. Adding a catalyst to a reaction at equilibrium:

a) Shifts it to the right b) Shifts it to the left c) Increases the yield d) Does not change the position of equilibrium

11. Which of these is an alkene?

a) C_2H_4 b) C_2H_6 c) C_2H_2 d) CH_4

12. Temporary hardness of water is caused by:

a) Calcium hydrogencarbonate b) Sodium chloride c) Calcium sulphate d) Magnesium chloride

13. The gas that bleaches damp litmus paper is:

a) Chlorine b) Nitrogen c) Hydrogen d) Carbon monoxide

14. The main ore of aluminium is:

a) Haematite b) Bauxite c) Galena d) Cassiterite

15. Ethanol and ethanoic acid react to form an ester. This reaction is called:

a) Saponification b) Hydrolysis c) Esterification d) Cracking

No.	Answer	No.	Answer
1	b) Fractional distillation	9	c) Cathode
2	c) 0.5	10	d) No change in position
3	d) 11.2 dm^3	11	a) C_2H_4
4	a) Group 1, period 3	12	a) Calcium hydrogencarbonate
5	c) Ionic	13	a) Chlorine
6	d) Carbon dioxide	14	b) Bauxite
7	a) Strongly alkaline	15	c) Esterification
8	b) Oxidised		

Recommended textbooks

- **New School Chemistry for Senior Secondary Schools**, Osei Yaw Ababio.
- **Senior Secondary Chemistry**, S. T. Bajah, B. O. Teibo, G. Onwu and A. Obikwere.
- Other texts listed in the current JAMB Chemistry syllabus.
- **Past questions**: solve at least the last ten years' UTME Chemistry questions, timing yourself, and study every one you get wrong.

Prepare with the Himaayah Academy

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وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ

"And my success is only through Allah." (Hūd 11:88)

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Master the mole, and the rest follows.

This guide from the Himaayah Academy helps UTME candidates prepare for Chemistry with a clear plan, the calculations that come every year and the facts to memorise.

- The JAMB Chemistry syllabus in six areas
- Mole, gas law and electrolysis formulas
- Tests for gases and ions, and organic series
- Nine worked examples, step by step
- An 8-week plan and a 15-question practice test

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